



US 20250281996A1

(19) **United States**(12) **Patent Application Publication**
YAMAGISHI et al.(10) **Pub. No.: US 2025/0281996 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **WELDED JOINT, WELDED MEMBER,
METHOD OF PRODUCING SAME, AND
RESISTANCE SPOT WELDING METHOD**(30) **Foreign Application Priority Data**

Jun. 3, 2022 (JP) 2022-091184

(71) Applicant: **JFE STEEL CORPORATION,**
Chiyoda-ku, Tokyo (JP)**Publication Classification**(72) Inventors: **Daiki YAMAGISHI,** Chiyoda-ku,
Tokyo (JP); **Chikaumi SAWANISHI,**
Chiyoda-ku, Tokyo (JP); **Nao**
KAWABE, Chiyoda-ku, Tokyo (JP);
Katsutoshi TAKASHIMA,
Chiyoda-ku, Tokyo (JP); **Koichi**
TANIGUCHI, Chiyoda-ku, Tokyo (JP);
Hiroshi MATSUDA, Chiyoda-ku,
Tokyo (JP)(51) **Int. Cl.****B23K 11/11** (2006.01)**B23K 101/18** (2006.01)**B23K 103/04** (2006.01)(52) **U.S. Cl.**CPC **B23K 11/115** (2013.01); **B23K 2101/18**
(2018.08); **B23K 2103/04** (2018.08)(73) Assignee: **JFE STEEL CORPORATION,**
Chiyoda-ku, Tokyo (JP)(21) Appl. No.: **18/858,748**(22) PCT Filed: **Jan. 27, 2023**(86) PCT No.: **PCT/JP2023/002727**

§ 371 (c)(1),

(2) Date: **Oct. 22, 2024**(57) **ABSTRACT**

Provided is a welded joint for which cracking of the welded portion is suppressed and nugget diameter is a desired size, even when at least one of the surface layer steel sheets of a sheet combination is a galvanized steel sheet and the thickness ratio of a surface layer steel sheet is large or a disturbance has a large effect. An indentation ratio b/T of shoulders of the sheet combination satisfies at least one of the following Expressions (1) or (2).

$$b/T \leq 0.6/(T/t_L)^{0.5} \times (980/S_L)^{0.5} \quad (1)$$

$$b/T \leq 0.6/(T/t_L)^{0.5} \times (980/S_L)^{0.5} \quad (2)$$

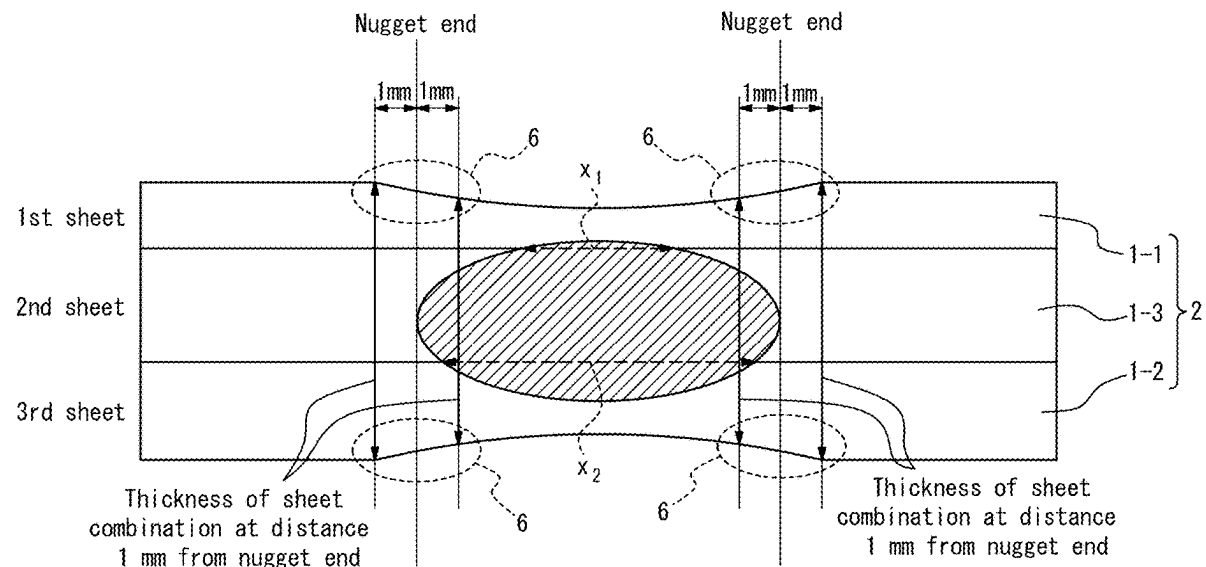


FIG. 1

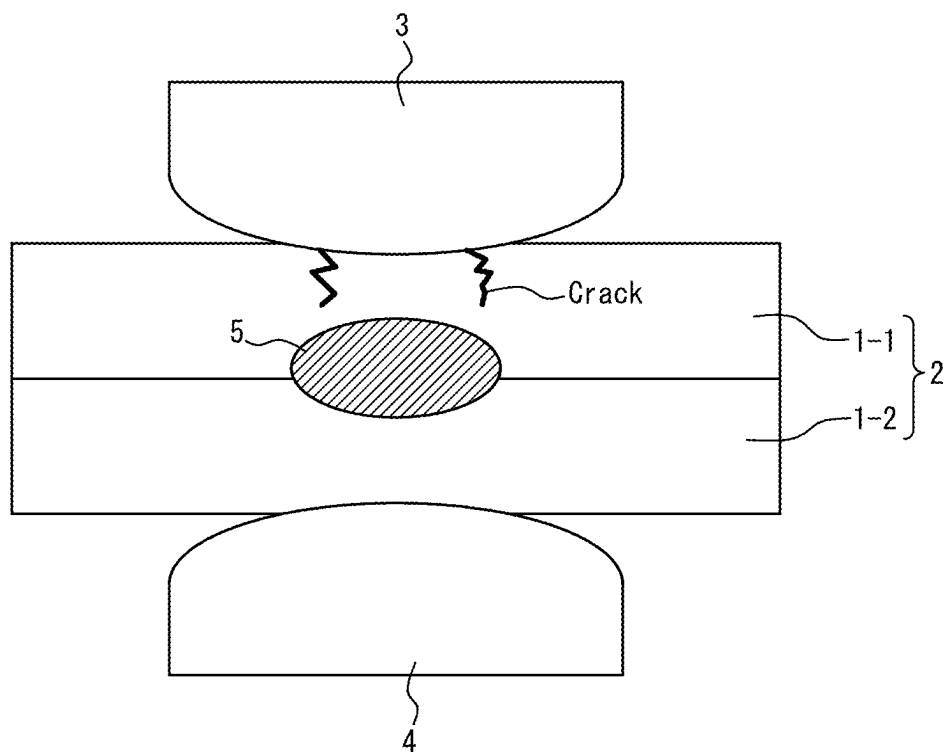


FIG. 2

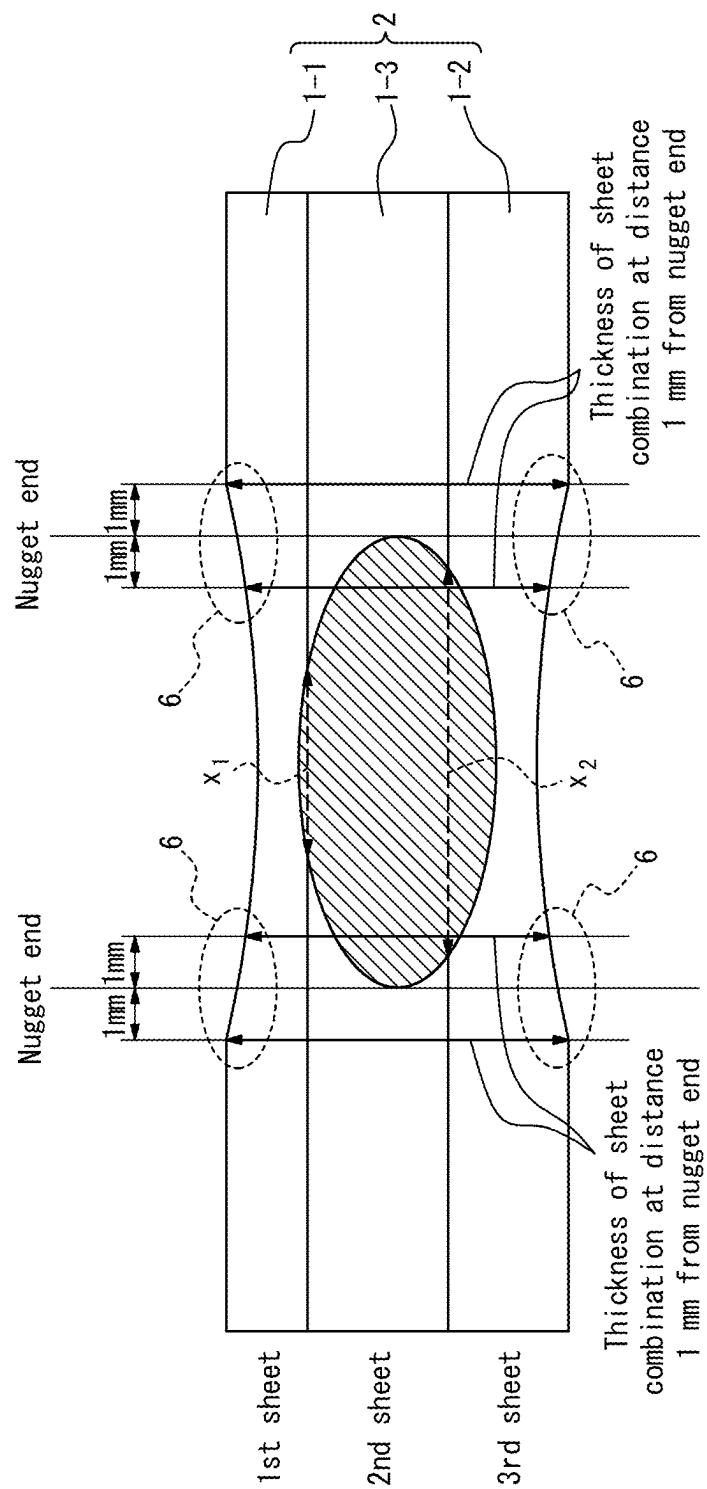
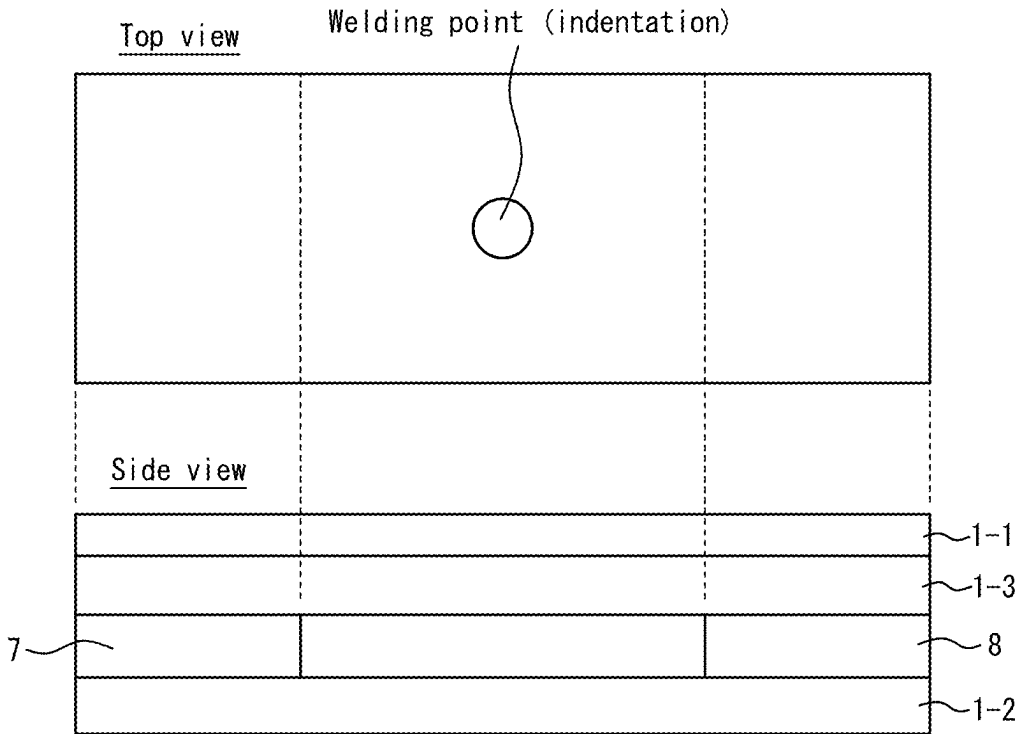


FIG. 3



**WELDED JOINT, WELDED MEMBER,
METHOD OF PRODUCING SAME, AND
RESISTANCE SPOT WELDING METHOD**

TECHNICAL FIELD

[0001] The present disclosure relates to a welded joint, a welded member, a method of producing same, and a resistance spot welding method.

BACKGROUND

[0002] In the assembly of automobiles, resistance spot welding, a type of lap resistance welding, is often used to join overlapping steel sheets from the viewpoint of cost and manufacturing efficiency. As illustrated in FIG. 1, for example, this welding method uses a pair of welding electrodes 3, 4 (hereinafter, the electrode disposed on the vertically upward side is also referred to as the upper electrode 3 and the electrode disposed on the vertically downward side is also referred to as the lower electrode 4) to squeeze a sheet combination 2 of overlapped steel sheets 1-1, 1-2 (hereinafter, the steel sheet disposed uppermost in the vertical direction is also referred to as the upper steel sheet 1-1 and the steel sheet disposed lowermost in the vertical direction is also referred to as the lower steel sheet 1-2). Pressure is applied from the welding electrodes 3, 4 from above and below while a welding current is passed between the welding electrodes 3, 4 to join the sheet combination 2. In this welding method, a point-like welded portion is obtained by using resistance heat generated by the welding current. The welded portion is called a nugget 5. That is, the nugget 5 is the portion that melts and solidifies at the contact point of the overlapped steel sheets when the electric current flows through the steel sheets. The steel sheets are spot-welded by the nugget 5.

[0003] In recent years, the automobile field has been promoting weight reduction of automotive bodies to improve fuel efficiency. As a result, the application of high strength steel sheets to automotive parts is increasing. Further, among automotive parts, for parts applied to locations that are exposed to rainwater, coated or plated steel sheets having antirust properties such as galvanized steel sheets (surface-treated steel sheets) are used from the viewpoint of corrosion resistance. Here, a coated or plated steel sheet is a steel sheet having a metal coating or plating layer on a surface of a base metal (base steel sheet). As a metal coating or plating layer, examples include zinc coating or plating as typified by electrogalvanized plating and hot-dip galvanizing (including galvannealing), zinc alloy coating or plating containing elements such as aluminum and magnesium in addition to zinc, and aluminum-zinc alloy coating or plating consisting mainly of aluminum and zinc, and the like.

[0004] As resistance spot welding applied to a sheet combination of an overlapped plurality of steel sheets including such a coated or plated steel sheet, an example is described in Patent Literature (PTL) 1:

[0005] “in spot welding of high strength coated or plated steel sheets, a high strength coated or plated steel sheet spot welding method in which the weld time and the hold time after welding current are set to satisfy the following conditions (1) and (2) to perform spot welding,

$$0.25 \cdot (10 \cdot t + 2)/50 \leq WT \leq 0.50 \cdot (10 \cdot t + 2)/50 \quad (1)$$

$$300 - 500 \cdot t + 250 \cdot t^2 \leq HT \quad (2)$$

[0006] where t is sheet thickness (mm), WT is weld time (ms), and HT is hold time (ms) after welding current.”

[0007] PTL 2 describes:

[0008] “in spot welding of high tensile strength galvanized steel sheets by multi-stage current having three or more stages, a high tensile strength galvanized steel sheet spot welding method in which welding conditions are adjusted so that the nugget formed is greater than or equal to a desired nugget diameter d_0 defined by the following expression (1) and a remaining thickness other than the nugget is 0.05 mm or more,

$$d_0 = k\sqrt{t} \quad (1)$$

[0009] where d_0 is the desired nugget diameter (mm),

[0010] k is a coefficient; a coefficient selected from 3 to 6 according to work conditions, and

[0011] t is steel sheet thickness (mm).”

[0012] PTL 3 describes:

[0013] “a spot welding method of spot welding a plurality of overlapped steel sheets by squeezing between opposed welding electrodes, the plurality of overlapped steel sheets including one or more steel sheets coated with a coating at the welding point on at least one surface, the method comprising

[0014] a process of removing the coating prior to spot welding,

[0015] wherein, in the process of removing the coating, a range where the coating is removed is at least an area that has an outer circumference in a circle that includes the outer edge of the heat-affected zone formed on the welding electrode side of the plurality of overlapped steel sheets.”

[0016] PTL 4 describes:

[0017] “a spot welding method of performing an actual welding process in which a member to be welded consisting of a plurality of steel sheets overlapped at least at the welding point is pressed and has a current passed therethrough by welding electrodes, and further, a current process of performing at least one of pre-current and subsequent-current,

[0018] wherein, for at least one of the plurality of steel sheets, at least an overlapping surface at the welding point is coated with a galvanized coating, the total thickness t (mm) of the plurality of steel sheets is 1.35 mm or more,

[0019] the pressure applied to the member to be welded by the welding electrodes is maintained from the start of current passing between the welding electrodes until the end of current passing between the welding electrodes at the end of welding,

[0020] and, at the end of welding, a hold time Ht (s) after the end of current passing between the welding electrodes until the welding electrodes and the member to be welded are no longer in contact is within the range of expression (1) below.

$$0.015t^2 + 0.020 \leq Ht \leq 0.16t^2 - 0.40t + 0.70'' \quad (1)$$

[0021] PTL 5 describes:

[0022] “a resistance spot welding method of squeezing, by a pair of welding electrodes, a sheet combination of a plurality of overlapped steel sheets and passing a current while applying an electrode force to join the sheet combination,

[0023] wherein at least one of the plurality of overlapped steel sheets is a surface-treated steel sheet with a metal coating or plating layer on a surface, the resistance spot welding method comprising:

[0024] a main current process of passing current to form a nugget;

[0025] a no-current process of suspending current after the main current process for a cooling time T_c (cycle); and

[0026] a subsequent-current process after the no-current process of passing current that reheats without growing the nugget,

[0027] wherein, when the inclined angle of the electrode is A (degrees), the current value of the main current process is I_m (kA), the current value of the subsequent-current process is I_p (kA), $1+0.1 \cdot T_c$ is a variable B , and $1+0.2 \cdot T_c$ is a variable C , the current satisfies the relationships in the following Expression (I),

when $0 < A < 3$, $(22 + A) \cdot B/100 < I_p/I_m < C$ Expression (I)

when $3 \leq A < 7$, $(17 + A) \cdot B/80 < I_p/I_m < C$

when $7 \leq A < 15$, $(11 + A) \cdot B/60 < I_p/I_m < C''$

CITATION LIST

Patent Literature

- [0028] PTL 1: JP 2003-103377 A
- [0029] PTL 2: JP 2003-236676 A
- [0030] PTL 3: WO 2016/159169 A1
- [0031] PTL 4: JP 2017-047476 A
- [0032] PTL 5: WO 2018/159764 A1

SUMMARY

Technical Problem

[0033] In resistance spot welding applied to a coated or plated steel sheet, in particular applied to a sheet combination consisting of a plurality of overlapped steel sheets including a galvanized steel sheet (hereinafter also referred to as resistance spot welding of a galvanized steel sheet), there is a problem that cracks easily occur in the welded portion.

[0034] Typically, the melting point of zinc or zinc alloy coating or plating is lower than that of base metal. Therefore, in resistance spot welding of a galvanized steel sheet, the metal coating or plating layer with a low melting point on the steel sheet surface melts during welding. When tensile stress due to the electrode pressing force and thermal expansion and contraction of the steel sheet is applied to the

welded portion, the molten low-melting-point metal penetrates into crystal grain boundaries of the base metal of the galvanized steel sheet, decreasing grain boundary strength and causing cracking. That is, most cracks in the welded portion that occur in resistance spot welding of a galvanized steel sheet are considered to be cracks caused by liquid metal embrittlement (hereinafter also referred to as liquid metal embrittlement cracks).

[0035] Such liquid metal embrittlement cracks are likely to occur when the welded portion is subjected to a large deformation. For example, when welding under conditions that cause expulsion (splashing), cracks are likely to occur on a surface of sheet combination 2 in contact with welding electrodes 3, 4, as illustrated in FIG. 1. In particular, a crack that occurs at a shoulder of a sheet combination surface, which is a periphery of a contact area between a welding electrode and a sheet combination (hereinafter also referred to as a shoulder crack), occurs from lower welding current values than a crack that occurs directly under an electrode.

[0036] On one hand, from the viewpoint of securing joint strength, securing a nugget of a defined size or larger is important. However, the larger the thickness ratio of an outermost steel sheet in a sheet combination (hereinafter also referred to as a surface layer steel sheet; in contact with an electrode during welding), the more difficult it becomes to secure a nugget diameter of a desired size between the surface layer steel sheet and a steel sheet adjacent to the surface layer steel sheet. Here, the thickness ratio of the surface layer steel sheet is a value obtained by dividing the total thickness of steel sheets of the sheet combination by the thickness of the surface layer steel sheet. Further, in implementation during automobile assembly, disturbances may occur during welding. For example, there may be a gap between a fixed welding electrode and a sheet combination, or between steel sheets of a sheet combination. In such cases, it also becomes difficult to secure a nugget diameter of the desired size.

[0037] In this regard, nugget diameter can be increased by setting a larger current value during welding. However, when the current value during welding is set to a large value, then, for example, there is an increased risk of expulsion causing large deformation of a welded portion, which increases the risk of the welded portion cracking, in particular increasing the risk of liquid metal embrittlement cracking, as typified by shoulder cracking.

[0038] The techniques described in PTL 1, 2, 4, and 5 are unable to both suppress the occurrence of cracks in the welded portion and secure a stable nugget diameter of the desired size when a thickness ratio of the surface layer steel sheet is large or a disturbance has a large effect, and improvement in this regard is desired at present. Improvement in this regard is desired not only for steel sheets for automobile use, but also for resistance spot welding of a galvanized steel sheet in other steel sheet applications.

[0039] Further, the technique of PTL 3 requires a process to remove the coated or plated layer from the steel sheet surface in advance (hereinafter also referred to as a coated or plated layer removal process), and therefore significantly increases manufacturing costs. Further, the coated or plated layer is removed from the steel sheet, which may lead to a decrease in corrosion resistance of the welded portion.

[0040] The present disclosure was developed in view of the current situation described above, and it would be helpful to provide a welded joint for which cracking of the

welded portion is suppressed and nugget diameter is a desired size, even when at least one of the surface layer steel sheets of a sheet combination is a galvanized steel sheet and the thickness ratio of a surface layer steel sheet is large or a disturbance has a large effect.

[0041] Further, it would be helpful to provide a resistance spot welding method for resistance spot welding of a galvanized steel sheet that does not require a coated or plated layer removal process and that can both suppress cracking of the welded portion and secure a stable nugget diameter of a desired size, even when the thickness ratio of a surface layer steel sheet is large or a disturbance has a large effect.

[0042] Further, it would be helpful to provide a welded member including the welded joint and a method of producing same.

Solution to Problem

[0043] The inventors engaged in extensive studies and made the following discoveries.

[0044] The presence or absence of welded portion cracking, in particular shoulder cracking, in resistance spot welding of a galvanized steel sheet, is strongly influenced by deformation at the shoulder of a sheet combination that occurs during welding, that is, a shoulder indentation ratio b/T . The shoulder indentation ratio b/T is appropriately controlled in relation to the tensile strength and thickness of the galvanized steel sheet arranged as the surface layer steel sheet of a sheet combination (the outermost steel sheet in the sheet combination (in contact with the electrode during welding)), and the total thickness of the steel sheets of the sheet combination. Specifically, at least one of Expression (1) or Expression (2) below is satisfied. This effectively suppresses the occurrence of cracking of the welded portion.

[0045] Here, the shoulder indentation ratio b/T is the value obtained by dividing the shoulder indentation amount b (mm) by the total thickness T (mm) of the steel sheets of the sheet combination. Further, the shoulder indentation amount b (mm) is the value obtained by subtracting the minimum thickness a (mm) of the sheet combination at a distance of 1 mm from the nugget end from the total thickness T (mm) of the steel sheets of the sheet combination.

[0046] Further, expulsion is one of the causes of deformation of the shoulder of the sheet combination that occurs during welding. By controlling the occurrence of expulsion (the amount of spattered molten metal), cracking of the welded portion can be suppressed. Based on this point, the inventors conducted further studies to obtain a resistance spot welding method that can both suppress cracking of the welded portion and secure a stable nugget diameter of the desired size, even when the thickness ratio of the surface layer steel sheet is large or a disturbance has a large effect.

[0047] As a result, the inventors found that it is effective to simultaneously satisfy the following (a) to (c).

[0048] (a) The current passing process during welding is divided into two processes: a first current process and a second current process.

[0049] (b) In the first current process, the amount of heat input is controlled by adjusting the current pattern to form a nugget while minimizing large deformation at the shoulders of the sheet combination.

[0050] (c) In the second current process, a current pattern of repeated current passing and cooling is used to gradually enlarge the nugget while minimizing large deformation at the shoulders of the sheet combination.

[0051] In the first current process, the nugget needs to be formed so that no large deformation occurs at the shoulders of the sheet combination in the second current process, which is a subsequent process. For this purpose, it is important to control the amount of heat input by adjusting the current pattern. In particular, regarding weld time α (ms) of the first current process, it is important to satisfy at least one of Expressions (4) or (5) below, depending on the thickness of the galvanized steel sheet disposed as the surface layer steel sheet of the sheet combination and the total thickness of the steel sheets of the sheet combination. This allows the pressure-welded portion around the nugget (corona bond) to be firmly joined. As a result, even when the thickness ratio of the surface layer steel sheet or a disturbance has a large effect, the amount of spattering expulsion, or excessive indentation, can be suppressed in the first current process while the nugget is enlarged. This makes it possible to suppress deformation of the shoulders of the sheet combination and therefore suppress cracking in the welded portion after the second current process is completed, while achieving the desired nugget diameter.

[0052] Further, in the second current process, current greater than or equal to the current value at the end of the first current process can be used to enlarge the nugget. However, when heat input is excessive, large deformation occurs at the shoulders of the sheet combination. Therefore, it is important to use a current pattern that repeats cooling and current passing for at least a certain period of time. Further, regarding a ratio β/γ of total cooling time β to total weld time in the second current process, it is important to satisfy at least one of Expressions (6) or (7) below, depending on the thickness of the galvanized steel sheet disposed as the surface layer steel sheet of the sheet combination and the total thickness of the steel sheets of the sheet combination. These allow the nugget to be enlarged gradually while minimizing large deformation at the shoulders of the sheet combination. The pressure-welded portion (corona bond) around the nugget is firmly joined in the first current process. Further, during the cooling in the second current process and the subsequent heating due to current passage, the pressure-welded portion around the nugget (corona bond) is more firmly joined. Therefore, even when expulsion occurs, the amount of expulsion (the amount of molten metal spattered) can be suppressed. As a result, deformation of the shoulders of the sheet combination is suppressed, and consequently cracking of the welded portion is suppressed.

[0053] In addition, the second current process enlarges the nugget as described above, and therefore the nugget is in a molten state at the end of the second current process. Therefore, when releasing the electrode from the sheet combination immediately after the second current passage process, cracks may occur from the nugget end or corona bond end between steel sheets of the sheet combination toward the inside of the nugget, resulting in a decrease in joint quality. Such cracks are more pronounced in a sheet combination of material to be joined where a boundary between the steel sheets of the sheet combination is located away from the center position ($1/2$ thickness position of the sheet combination) of the overall sheet combination thickness (total thickness of the steel sheets of the sheet combination). Examples of such sheet combinations include sheet combinations that consist of two steel sheets of different thicknesses stacked on top of each other, and sheet combinations that consist of three or more steel sheets stacked on

top of each other, regardless of their thickness. The inventors have also studied this point and found that the crack initiation described above can be effectively prevented regardless of sheet combination by providing a pressure hold process in which pressure is held in a no-current state after the second current process and by setting the pressure hold time to 10 ms or more in the pressure hold process.

[0054] The present disclosure is based on these discoveries and further studies.

[0055] Primary features of the present disclosure are as follows.

[0056] 1. A welded joint comprising a sheet combination of n overlapped steel sheets and a nugget joining the steel sheets, wherein

[0057] n is an integer greater than or equal to 2,

[0058] in the sheet combination, at least one of the first steel sheet or the nth steel sheet in order from the top is a galvanized steel sheet,

[0059] an indentation ratio b/T of shoulders of the sheet combination satisfies,

[0060] in the case of [Condition 1], the following Expression (1),

[0061] in the case of [Condition 2], the following Expression (2),

[0062] in the case of [Condition 3], Expressions (1) and (2),

[0063] in the sheet combination, nugget diameter x_k , in mm, at each boundary level between a kth steel sheet and a (k+1)th steel sheet is $3.5\sqrt{t_k}$ or more, k is an integer from 1 to n-1, t_k is the thickness in mm of the thinner of the kth steel sheet and the (k+1)th steel sheet,

$$b/T \leq 0.6/(T/t_U)^{0.5} \times (980/S_U)^{0.5} \quad (1)$$

$$b/T \leq 0.6/(T/t_L)^{0.5} \times (980/S_L)^{0.5} \quad (2)$$

[0064] where

[0065] b is shoulder indentation in mm,

[0066] T is total thickness in mm of the n steel sheets,

[0067] t_U is thickness in mm of the first steel sheet,

[0068] t_L is thickness in mm of the nth steel sheet,

[0069] S_U is tensile strength in MPa of the first steel sheet,

[0070] S_L is tensile strength in MPa of the nth steel sheet,

[0071] the shoulder indentation b is determined by the following Expression (3),

$$b = T - a \quad (3)$$

[0072] where

[0073] a is minimum thickness in mm of the sheet combination at a distance of 1 mm from a nugget end,

[0074] and [Condition 1] to [Condition 3] are as follows:

[0075] [Condition 1]

[0076] of the first steel sheet and the nth steel sheet of the sheet combination, only the first steel sheet is a galvanized steel sheet,

[0077] [Condition 2]

[0078] of the first steel sheet and the nth steel sheet of the sheet combination, only the nth steel sheet is a galvanized steel sheet,

[0079] [Condition 3]

[0080] of the first steel sheet and the nth steel sheet of the sheet combination, both the first steel sheet and the nth steel sheet are galvanized steel sheets.

[0081] 2. A welded member comprising the welded joint according to 1, above.

[0082] 3. A resistance spot welding method of squeezing, by a pair of welding electrodes, a sheet combination of n overlapped steel sheets and passing a current while applying an electrode force to join the sheet combination, wherein

[0083] n is an integer greater than or equal to 2,

[0084] in the sheet combination, at least one of the first steel sheet or the nth steel sheet in order from the top is a galvanized steel sheet,

[0085] the resistance spot welding method comprising:

[0086] a first current process of forming a nugget;

[0087] a second current process of enlarging the nugget formed in the first current passage process; and

[0088] after the second current process, a pressure hold process of holding pressure on the sheet combination in a no-current state,

[0089] wherein,

[0090] in the first current process, current is passed for a weld time of α ms,

[0091] the weld time α , in ms, satisfies,

[0092] in the case of [Condition 1], the following Expression (4),

[0093] in the case of [Condition 2], the following Expression (5),

[0094] in the case of [Condition 3], Expressions (4) and (5),

[0095] the second current process comprises:

[0096] cooling where no current is passed for a cooling time of 10 ms or more; and

[0097] current passing for a weld time of 15 ms or more with a current value greater than or equal to a current value of the first current process, the cooling and the current passing each being carried out at least once, wherein

[0098] a ratio β/γ of total cooling time β to total weld time γ in the second current process satisfies,

[0099] in the case of [Condition 1], Expression (6),

[0100] in the case of [Condition 2], Expression (7),

[0101] in the case of [Condition 3], Expressions (6) and (7),

[0102] in the pressure hold process, the pressure hold time is 10 ms or more,

$$\alpha \geq 50 \times (T/t_U)^{0.5} \quad (4)$$

$$\alpha \geq 50 \times (T/t_L)^{0.5} \quad (5)$$

$$0.10 \times (T/t_U)^{0.5} \leq \beta/\gamma \leq 2 \times (T/t_U)^{0.5} \quad (6)$$

$$0.10 \times (T/t_L)^{0.5} \leq \beta/\gamma \leq 2 \times (T/t_L)^{0.5} \quad (7)$$

[0103] where

[0104] T is total thickness in mm of the n steel sheets,

[0105] t_U is thickness in mm of the first steel sheet,

[0106] t_L is thickness in mm of the nth steel sheet,

[0107] and [Condition 1] to [Condition 3] are as follows:

[0108] [Condition 1]

[0109] of the first steel sheet and the nth steel sheet of the sheet combination, only the first steel sheet is a galvanized steel sheet,

- [0110] [Condition 2]
- [0111] of the first steel sheet and the nth steel sheet of the sheet combination, only the nth steel sheet is a galvanized steel sheet,
- [0112] [Condition 3]
- [0113] of the first steel sheet and the nth steel sheet of the sheet combination, both the first steel sheet and the nth steel sheet are galvanized steel sheets.
- [0114] 4. The resistance spot welding method according to 3, above, satisfying one or more of the following conditions (A) to (E),
- [0115] (A) a welding electrode has an inclined angle relative to the sheet combination,
- [0116] (B) the pair of welding electrodes are off-center,
- [0117] (C) before pressing the sheet combination, there is a gap between the fixed welding electrode and the sheet combination,
- [0118] (D) before pressing the sheet combination, there is at least one gap between the steel sheets of the sheet combination, and
- [0119] (E) on the surface of the sheet combination, the shortest distance from the center of the welding contact point to an end face of the sheet combination is 10 mm or less.
- [0120] 5. A method of producing a welded member, comprising a process of joining a sheet combination of n overlapped steel sheets by the resistance spot welding method according to 3 or 4, above, wherein n is an integer greater than or equal to 2.

Advantageous Effect

[0121] According to the present disclosure, a welded joint is obtainable for which cracking of the welded portion is suppressed and nugget diameter is a desired size, even when at least one of the surface layer steel sheets of a sheet combination is a galvanized steel sheet and the thickness ratio of a surface layer steel sheet is large or a disturbance has a large effect.

[0122] Further, a welded member including the welded joint of the present disclosure includes a galvanized steel sheet having high corrosion resistance disposed outermost, and is therefore extremely suitable for application to automotive parts and the like, in particular automotive parts used in locations that are exposed to rainwater.

BRIEF DESCRIPTION OF THE DRAWINGS

- [0123] In the accompanying drawings:
- [0124] FIG. 1 is a diagram schematically illustrating an example resistance spot welding method;
- [0125] FIG. 2 is a diagram schematically illustrating an example of a welded joint cross-section; and
- [0126] FIG. 3 is a diagram schematically illustrating an example of a sheet combination that has a sheet gap.

DETAILED DESCRIPTION

[0127] The following describes embodiments of the present disclosure. First, a welded joint according to an embodiment of the present disclosure is described.

[1] Welded Joint

[0128] The welded joint according to an embodiment of the present disclosure is

[0129] a welded joint comprising a sheet combination of n overlapped steel sheets and a nugget joining the steel sheets, wherein

[0130] n is an integer greater than or equal to 2,

[0131] in the sheet combination, at least one of the first steel sheet or the nth steel sheet in order from the top is a galvanized steel sheet,

[0132] an indentation ratio b/T of shoulders of the sheet combination satisfies,

[0133] in the case of [Condition 1], the following Expression (1),

[0134] in the case of [Condition 2], the following Expression (2),

[0135] in the case of [Condition 3], Expressions (1) and (2),

[0136] in the sheet combination, nugget diameter x_k , in mm, at each boundary level between a kth steel sheet and a (k+1)th steel sheet is $3.5\sqrt{t_k}$ or more, k is an integer from 1 to n-1, and t_k is the thickness in mm of the thinner of the kth steel sheet and the (k+1)th steel sheet.

$$b/T \leq 0.6/(T/t_U)^{0.5} \times (980/S_U)^{0.5} \quad (1)$$

$$b/T \leq 0.6/(T/t_L)^{0.5} \times (980/S_L)^{0.5} \quad (2)$$

[0137] Here,

[0138] b is shoulder indentation in mm,

[0139] T is total thickness in mm of the n steel sheets,

[0140] t_U is thickness in mm of the first steel sheet,

[0141] t_L is thickness in mm of the nth steel sheet,

[0142] S_U is tensile strength in MPa of the first steel sheet, and

[0143] S_L is tensile strength in MPa of the nth steel sheet.

[0144] Further, the shoulder indentation b is determined by the following Expression (3).

$$b = T - a \quad (3)$$

[0145] Here,

[0146] a is minimum thickness in mm of the sheet combination at a distance of 1 mm from a nugget end.

[0147] In addition, [Condition 1] to [Condition 3] are as follows.

[Condition 1]

[0148] Of the first steel sheet and the nth steel sheet of the sheet combination, only the first steel sheet is a galvanized steel sheet.

[Condition 2]

[0149] Of the first steel sheet and the nth steel sheet of the sheet combination, only the nth steel sheet is a galvanized steel sheet.

[Condition 3]

[0150] Of the first steel sheet and the nth steel sheet of the sheet combination, both the first steel sheet and the nth steel sheet are galvanized steel sheets.

[0151] Here, the sheet combination is n steel sheets stacked on top of each other, and at least one of the first steel sheet and the n th steel sheet in order from the top, that is, at least one of the surface layer steel sheets of the sheet combination, is a galvanized steel sheet. The order from the top here is, for example, the order from the top in the vertical direction when the sheet combination is arranged so that surfaces of the sheet combination are parallel to the horizontal plane.

[0152] Further, a galvanized steel sheet is a steel sheet that has a zinc or zinc alloy coated or plated layer on one or both surfaces of the base steel sheet, such as a hot-dip galvanized steel sheet or a galvanized steel sheet. When a steel sheet with a zinc or zinc alloy coated or plated layer on one surface of the base steel sheet is used as the first steel sheet or the n th steel sheet, the zinc or zinc alloy coated or plated layer is preferably outermost in the sheet combination. Here, the zinc or zinc alloy coated or plated layer is a coated or plated layer having a zinc content of 1 mass % or more. The zinc content is preferably 30% or more. The term zinc or zinc alloy coated or plated layer includes, for example, a hot-dip galvanized layer, a galvanized layer, an electrogalvanized layer, as well as a zinc alloy coated or plated layer containing a total of less than 50 mass % of alloying elements such as aluminum, magnesium, silicon, nickel, and iron. Further, as (a steel sheet including) a zinc alloy coated or plated layer, examples include Galfan (Zn-5 mass % Al) and Ecogal® (Ecogal is a registered trademark in Japan, other countries, or both) (Zn-5 mass % Al-1 mass % or less Mg and Ni). Further, as a zinc alloy coated or plated layer, examples include an aluminum-zinc alloy coated or plated layer that contains a total of 67 mass % or more of aluminum and zinc and less than 33 mass % of alloying elements such as magnesium, silicon, nickel, and iron (for example, Galvalume (55 mass % Al-43.4 mass % Zn-1.6 mass % Si), and the like). The zinc or zinc alloy coated or plated layer preferably has a melting point lower than that of the base steel sheet. Further, the balance other than the zinc and alloying elements described above is inevitable impurity. Further, there is no particular limitation for the base steel sheet. For example, steel sheets having various strengths can be applied, from mild steel having 270 MPa grade tensile strength (hereinafter also referred to as TS) to a steel sheet having 490 MPa grade to 2500 MPa grade TS.

[0153] In addition, a galvanized steel sheet as described above may be used for a steel sheet other than the first steel sheet and the n th steel sheet out of the steel sheets of the sheet combination. Further, a steel sheet without coating or plating may be used. For example, steel sheets having various strengths may be used, from mild steel having 270 MPa grade TS to a steel sheet having 490 MPa grade to 2500 MPa grade TS. When the first steel sheet is a galvanized steel sheet, the n th steel sheet may be a galvanized steel sheet as described above, or a steel sheet without coating or plating as described above. The same is true for the first steel sheet when the n th steel sheet is a galvanized steel sheet.

[0154] The thickness of each of the steel sheets of the sheet combination is not particularly limited. For example, thickness is preferably 0.4 mm or more. Thickness is preferably 3.2 mm or less. Steel sheets having a thickness of 0.4 mm or more to 3.2 mm or less are suitable for use as members for automobiles.

[0155] Further, the number of overlapped steel sheets in a sheet combination, n , is an integer greater than or equal to

2. An upper limit of n is not particularly limited. For example, n is preferably 7 or less.

[0156] In the welded joint according to an embodiment of the present disclosure, it is important to satisfy at least one of Expressions (1) or (2) for the shoulder indentation ratio b/T , depending on [Condition 1] to [Condition 3]. Here, shoulder refers to a shoulder at an indentation on the surface (front and back) of the sheet combination, as illustrated in FIG. 2. In the drawing, reference signs **1-1** to **1-3** indicate steel sheets and **6** indicates a shoulder. Further, the indentations on the surfaces of the sheet combination (front and back surfaces) are welding electrode marks caused by the pressure applied by the welding electrodes during welding, and the nugget is located between the indentations.

[0157] Shoulder indentation ratio b/T : satisfy at least one of Expressions (1) or (2), depending on [Condition 1] to [Condition 3].

[0158] As mentioned above, the presence or absence of welded portion cracking, in particular shoulder cracking, in resistance spot welding of a galvanized steel sheet, is strongly influenced by deformation at the shoulder of a sheet combination that occurs during welding, that is, the shoulder indentation ratio b/T . By appropriately controlling the shoulder indentation ratio b/T in relation to the tensile strength and thickness of a galvanized steel sheet arranged as the surface layer steel sheet of a sheet combination, and the total thickness of the steel sheets of the sheet combination, occurrence of welded portion cracking can be effectively suppressed.

[0159] Specifically, occurrence of welded portion cracking can be effectively suppressed by satisfying,

[0160] in the case of [Condition 1], the following Expression (1),

[0161] in the case of [Condition 2], the following Expression (2),

[0162] in the case of [Condition 3], Expressions (1) and (2).

$$b/T \leq 0.6/(T/t_U)^{0.5} \times (980/S_U)^{0.5} \quad (1)$$

$$b/T \leq 0.6/(T/t_L)^{0.5} \times (980/S_L)^{0.5} \quad (2)$$

[0163] Here,

[0164] b is shoulder indentation in mm,

[0165] T is total thickness in mm of the n steel sheets,

[0166] t_U is thickness in mm of the first steel sheet,

[0167] t_L is thickness in mm of the n th steel sheet,

[0168] S_U is tensile strength in MPa of the first steel sheet, and

[0169] S_L is tensile strength in MPa of the n th steel sheet.

[0170] Further, the shoulder indentation b is determined by the following Expression (3).

$$b = T - a \quad (3)$$

[0171] Here,

[0172] a is minimum thickness in mm of the sheet combination at a distance of 1 mm from a nugget end.

[0173] In addition, [Condition 1] to [Condition 3] are as follows.

[Condition 1]

[0174] Of the first steel sheet and the nth steel sheet of the sheet combination, only the first steel sheet is a galvanized steel sheet.

[Condition 2]

[0175] Of the first steel sheet and the nth steel sheet of the sheet combination, only the nth steel sheet is a galvanized steel sheet.

[Condition 3]

[0176] Of the first steel sheet and the nth steel sheet of the sheet combination, both the first steel sheet and the nth steel sheet are galvanized steel sheets.

[0177] Here, the minimum thickness of the sheet combination at a distance of 1 mm from the nugget end is measured as follows.

[0178] The welded joint is cut perpendicular to the surface of the sheet combination so as to pass through the center of the nugget. On the cross-section, as illustrated in FIG. 2, the thickness of the sheet combination is measured at a distance of 1 mm from the nugget end (the end in a direction perpendicular to the thickness direction of the sheet combination), and the minimum thickness is a (mm). The measurement of nugget diameter, described below, is also performed on the cross-section.

[0179] The shoulder indentation ratio b/T in Expression (1) is preferably $b/T \leq 0.4/(T/t_U)^{0.5} \times (980/S_U)^{0.5}$. The shoulder indentation ratio b/T in Expression (2) is preferably $b/T \leq 0.4/(T/t_L)^{0.5} \times (980/S_L)^{0.5}$. A lower limit of the shoulder indentation ratio b/T is not particularly limited and may be 0.

[0180] Further, as mentioned above, in a sheet combination of material to be joined where a boundary between steel sheets of the sheet combination is located away from the center position of the thickness of the entire sheet combination ($1/2$ thickness position of the sheet combination), that is, in a sheet combination where T/t_U or T/t_L is large, cracks are likely to occur from the nugget end or corona bond end between steel sheets of the sheet combination toward the inside of the nugget. Therefore, regarding T/t_U and T/t_L , the occurrence of such cracks is preferably suppressed in a sheet combination that satisfies at least one of the following Expressions (8) or (9). Further, the occurrence of such cracks is more preferably suppressed in a sheet combination that satisfies at least one of the following Expressions (10) or (11).

$$T/t_U > 2 \quad (8)$$

$$T/t_L > 2 \quad (9)$$

$$T/t_U > 3 \quad (10)$$

$$T/t_L > 3 \quad (11)$$

[0181] Nugget diameter x_k (mm) at boundary level between kth steel sheet and (k+1)th steel sheet: $3.5\sqrt{t_k}$ or more

[0182] From the viewpoint of ensuring joint strength, the nugget diameter x_k (mm) at each boundary level between the kth steel sheet and the (k+1)th steel sheet is $3.5\sqrt{t_k}$ or more.

The nugget diameter x_k is preferably $4.0\sqrt{t_k}$ or more. An upper limit of the nugget diameter x_k is not particularly limited. From the viewpoint of suppressing expulsion, the nugget diameter x_k is preferably $10.0\sqrt{t_k}$ or less. Here, k is an integer from 1 to n-1 and t_k is the thickness (mm) of the kth steel sheet or the (k+1)th steel sheet, whichever is thinner. The nugget is a point-like welded portion that joins the steel sheets of a sheet combination. Further, the nugget is a portion of the sheet combination where the steel sheets have melted and solidified.

[2] Welded Member

[0183] The welded member according to an embodiment of the present disclosure is a welded member including the welded joint described above. The welded member according to an embodiment of the present disclosure includes a galvanized steel sheet having high corrosion resistance disposed outermost, and is therefore suitable for application to automotive parts and the like, in particular automotive parts used in locations that are exposed to rainwater. The welded member according to an embodiment of the present disclosure may further include another welded joint (welded portion) in addition to the welded joint described above.

[3] Resistance Spot Welding Method

[0184] The resistance spot welding method according to an embodiment of the present disclosure is

[0185] a resistance spot welding method of squeezing, by a pair of welding electrodes, a sheet combination of n overlapped steel sheets and passing a current while applying an electrode force to join the sheet combination, wherein

[0186] n is an integer greater than or equal to 2,

[0187] in the sheet combination, at least one of the first steel sheet or the nth steel sheet in order from the top is a galvanized steel sheet,

[0188] the resistance spot welding method comprising:

[0189] a first current process of forming a nugget;

[0190] a second current process of enlarging the nugget formed in the first current passage process; and

[0191] after the second current process, a pressure hold process of holding pressure on the sheet combination in a no-current state,

[0192] wherein,

[0193] in the first current process, current is passed for a weld time of α ms,

[0194] the weld time α , in ms, satisfies,

[0195] in the case of [Condition 1], the following Expression (4),

[0196] in the case of [Condition 2], the following Expression (5),

[0197] in the case of [Condition 3], Expressions (4) and (5),

[0198] the second current process comprises:

[0199] cooling where no current is passed for a cooling time of 10 ms or more; and

[0200] current passing for a weld time of 15 ms or more with a current value greater than or equal to a current value of the first current process, the cooling and the current passing each being carried out at least once, wherein

[0201] a ratio β/γ of total cooling time β to total weld time γ in the second current process satisfies,

- [0202] in the case of [Condition 1], Expression (6),
 [0203] in the case of [Condition 2], Expression (7),
 [0204] in the case of [Condition 3], Expressions (6) and (7),
 [0205] and, in the pressure hold process, the pressure hold time is 10 ms or more.

$$\alpha \geq 50 \times (T/t_U)^{0.5} \quad (4)$$

$$\alpha \geq 50 \times (T/t_L)^{0.5} \quad (5)$$

$$0.10 \times (T/t_U)^{0.5} \leq \beta/\gamma \leq 2 \times (T/t_U)^{0.5} \quad (6)$$

$$0.10 \times (T/t_L)^{0.5} \leq \beta/\gamma \leq 2 \times (T/t_L)^{0.5} \quad (7)$$

[0206] Here,

[0207] T is total thickness in mm of the n steel sheets,

[0208] t_U is thickness in mm of the first steel sheet, and

[0209] t_L is thickness in mm of the nth steel sheet.

[0210] Further, [Condition 1] to [Condition 3] are as follows.

[Condition 1]

[0211] Of the first steel sheet and the nth steel sheet of the sheet combination, only the first steel sheet is a galvanized steel sheet.

[Condition 2]

[0212] Of the first steel sheet and the nth steel sheet of the sheet combination, only the nth steel sheet is a galvanized steel sheet.

[Condition 3]

[0213] Of the first steel sheet and the nth steel sheet of the sheet combination, both the first steel sheet and the nth steel sheet are galvanized steel sheets.

[0214] For example, as illustrated in FIG. 1, the pair of welding electrodes 3, 4 (the upper electrode 3 and the lower electrode 4) squeeze the sheet combination 2 of the overlapped steel sheets 1-1, 1-2 (also referred to as the upper steel sheet 1-1 and the lower steel sheet 1-2). Pressure is applied by the welding electrodes 3, 4 from above and below while a welding current is passed between the welding electrodes 3, 4 to join the sheet combination 2. Any welding device that includes a pair of upper and lower welding electrodes and is capable of freely controlling each of the electrode force and the welding current during welding may be used in the resistance spot welding method according to an embodiment of the present disclosure. Type (stationary, robot gun, and the like), electrode shape, and the like are not particularly limited. Further, a configuration for adding and controlling electrode force is not particularly limited, and a conventionally known device such as an air cylinder, a servo motor, and the like may be used. Further, a configuration for supplying current and controlling the current value during current passing is not particularly limited, and any conventionally known device may be used. Further, the current during current passing may be direct current or alternating current. In the case of alternating current, “current” means “effective current”. Further, type of tip of the upper electrode 3 and the lower electrode 4 is also not particularly limited. Examples include a dome-radius (DR) type, radius (R) type,

dome (D) type, and the like, as described in Japanese Industrial Standard JIS C 9304:1999. Further, tip diameter of each electrode is, for example, 4 mm to 16 mm.

[0215] In the resistance spot welding method according to an embodiment of the present disclosure, the current process during welding is divided into two processes, the first current process and the second current process, as described above. In the first current process, the amount of heat input is controlled by adjusting the current pattern to form the nugget while minimizing large deformation at the shoulders of the sheet combination. In the second current process, a current pattern of repeated current passing and cooling is used to gradually enlarge the nugget while minimizing large deformation at the shoulders of the sheet combination.

[0216] The first current process and the second current process are described below. The description of the sheet combination used in the resistance spot welding method according to an embodiment of the present disclosure is the same as the description of the sheet combination under [1] Welded joint, and is therefore omitted here.

First Current Process

[0217] In the first current process, the nugget needs to be formed so that no large deformation occurs at the shoulders of the sheet combination in the second current process, which is a subsequent process. For this purpose, it is important to control the amount of heat input by adjusting the current pattern, and in particular, to appropriately control the weld time.

[0218] Weld time α (ms): at least one of Expressions (4) or (5) is satisfied, depending on [Condition 1] to [Condition 3].

[0219] As mentioned above, in the first current process, the amount of heat input is controlled by adjusting the current pattern. In particular, regarding weld time α (ms), at least one of Expressions (4) or (5) is satisfied, depending on the thickness of the galvanized steel sheet disposed as the surface layer steel sheet of the sheet combination and the total thickness of the steel sheets of the sheet combination. This firmly joins the pressure-welded portion (corona bond) around the nugget. As a result, even when the thickness ratio of the surface layer steel sheet or a disturbance has a large effect, the amount of spattering expulsion, or excessive indentation, can be suppressed in the first current process while the nugget is enlarged. This makes it possible to suppress deformation of the shoulders of the sheet combination and therefore suppress cracking in the welded portion after the second current process is completed, while achieving the desired nugget diameter.

$$\alpha \geq 50 \times (T/t_U)^{0.5} \quad (4)$$

$$\alpha \geq 50 \times (T/t_L)^{0.5} \quad (5)$$

[0220] Here,

[0221] T is total thickness in mm of the n steel sheets,

[0222] t_U is thickness in mm of the first steel sheet, and

[0223] t_L is thickness in mm of the nth steel sheet.

[0224] Further, [Condition 1] to [Condition 3] are as described above.

[0225] The weld time α (ms) in Expression (4) is preferably $\alpha \geq 70 \times (T/t_U)^{0.5}$. The weld time α (ms) in Expression (5)

is preferably $\alpha \geq 70 \times (T/t_L)^{0.5}$. An upper limit of the weld time α (ms) is not particularly limited. From the viewpoint of production efficiency, the weld time α is preferably 1000 ms or less.

[0226] Conditions other than those described above are not particularly limited, and a conventional method may be used. For example, the current value for the first current process (hereinafter also simply referred to as I1) may be 2.0 kA to 15.0 kA, and the electrode force may be 1.5 kN to 10.0 kN. In the first current process, use of upslope current is preferred. For example, when the current value at the start of current passing is Is (kA) and the current value at the end of current passing is If (kA), the following Expression (12) is preferably satisfied. This allows the effect of firmly joining the pressure-welded portion (corona bond) around the nugget and the effect of forming a large nugget while suppressing deformation in the first current process to be more effectively obtained.

$$If > Is \quad (12)$$

[0227] Is is preferably 2.0 kA to 14.0 kA. If is preferably 3.0 kA to 15.0 kA.

[0228] Further, when the current value changes during current passing, the maximum current value in the first current process (that is, the current value at the end of current passing when upslope current is used, as described above) is considered as I1 (current value in the first current process).

Second Current Process

[0229] In the second current process, a current pattern of repeated cooling and current passing is used to gradually enlarge the nugget while minimizing large deformation at the shoulders of the sheet combination.

[0230] Cooling in the no-current state for cooling time: 10 ms or more, and current passing for weld time: 15 ms or more at current value: I1 or more, each at least once

[0231] As mentioned above, in the second current process, it is necessary to enlarge the nugget gradually while minimizing large deformation at the shoulders of the sheet combination. Therefore, in the second current process, the current pattern repeats cooling and current passing. In particular, cooling is performed for a cooling time of 10 ms or more, and current is passed for a weld time of 15 ms or more at a current value of I1 or more, each at least once. From the viewpoint of more advantageously obtaining the effects described above, the number of cycles of cooling and current passing is preferably two or more. When the number of cycles of cooling and current passing exceeds ten, the effects described above may become saturated, while at the same time, production efficiency may decrease. Therefore, the number of cycles of cooling and current passing is preferably ten or less.

[0232] Further, in the cooling, when the cooling time per cycle is less than 10 ms, the cooling effect is not sufficient and large deformation is likely to occur at the shoulders of the sheet combination, making cracking of the welded portion more likely. Therefore, the cooling time per cycle is 10 ms or more. From the viewpoint of enlarging the nugget gradually while more advantageously suppressing large deformation at the shoulders of the sheet combination, the

cooling time per cycle is preferably 15 ms or more. An upper limit of the cooling time per cycle is not particularly limited. From the viewpoint of production efficiency, the cooling time per cycle is preferably 300 ms or less. Cooling is performed as a no-current state, and non-welding time is the cooling time.

[0233] Further, in the current passing, when the weld time per cycle is less than 15 ms, the nugget cannot be sufficiently enlarged in the second current process due to insufficient heat input. Therefore, the weld time per cycle is 15 ms or more. From the viewpoint of enlarging the nugget gradually while more advantageously suppressing large deformation at the shoulders of the sheet combination, the weld time per cycle is preferably 20 ms or more. An upper limit of the weld time per cycle is not particularly limited. From the viewpoint of suppressing expulsion, the weld time per cycle is preferably 300 ms or less.

[0234] In addition, in the current passing described above, the current value needs to be I1 or more, that is, greater than or equal to the current value of the first current process, in order to enlarge the nugget gradually. The current value in the current passing described above is preferably $1.1 \times I1$ or more. An upper limit of the current value in the current passing described above is not particularly limited. From the viewpoint of suppressing the occurrence of expulsion, the current value in the current passing described above is preferably 20 kA or less.

[0235] The cooling time per cycle in the cooling may be the same or different for each cooling cycle, as long as the cooling time is 10 ms or more. The same applies to the weld time and current value in the current passing described above.

[0236] Ratio β/γ of the total cooling time β to the total weld time γ : at least one of Expressions (6) or (7) is satisfied, depending on [Condition 1] to [Condition 3].

[0237] In the resistance spot welding method according to an embodiment of the present disclosure, it is extremely important that the ratio β/γ of the total cooling time β to the total weld time γ in the second current process satisfies,

[0238] in the case of [Condition 1], Expression (6),

[0239] in the case of [Condition 2], Expression (7), or

[0240] in the case of [Condition 3], Expressions (6) and (7).

[0241] As a result, even when expulsion occurs in the second current process, the pressure-welded portion around the nugget (corona bond) is more firmly joined during cooling in the second current process and subsequent heating due to current passage. Therefore, the amount of expulsion (the amount of molten metal spattered) can be suppressed. As a result, it is possible to achieve a nugget diameter of the desired size, specifically, the nugget diameter x_k at each boundary level between the k th steel sheet and the $(k+1)$ th steel sheet of $3.5\sqrt{t_k}$ or more, while suppressing deformation at the shoulders of the sheet combination and thereby suppressing cracking in the welded portion.

$$0.10 \times (T/t_U)^{0.5} \leq \beta/\gamma \leq 2 \times (T/t_U)^{0.5} \quad (6)$$

$$0.10 \times (T/t_L)^{0.5} \leq \beta/\gamma \leq 2 \times (T/t_L)^{0.5} \quad (7)$$

[0242] Here,

[0243] T is total thickness in mm of the n steel sheets,

[0244] t_U is thickness in mm of the first steel sheet, and

[0245] t_L is thickness in mm of the nth steel sheet.

[0246] Further, [Condition 1] to [Condition 3] are as described above.

[0247] Here, when β/γ is too small, the effect of suppressing the amount of expulsion spattering due to cooling is not sufficient, and cracking in the welded portion is not suppressed. On the other hand, when β/γ is too large, the effect of nugget diameter enlargement due to current passage is not sufficient, and the desired nugget diameter cannot be obtained while suppressing cracking in the welded portion. β/γ in Expression (6) is preferably $0.14 \times (T/t_L)^{0.5} \leq \beta/\gamma \leq 1.6 \times (T/t_L)^{0.5}$. β/γ in Expression (7) is preferably $0.14 \times (T/t_L)^{0.5} \leq \beta/\gamma \leq 1.6 \times (T/t_L)^{0.5}$.

[0248] In the second current process, as mentioned above, the nugget formed in the first current process needs to be enlarged, and the amount of nugget enlargement is preferably 0.05 mm or more. Here, the amount of nugget enlargement in the second current process is calculated as the maximum value of $x_k - x_k'$. x_k' is the nugget diameter at each boundary level between the k th steel sheet and the $(k+1)$ th steel sheet at the end of the first current process. x_k' can be determined, for example, by separately preparing a sample in which current is passed up to the first current process under the same conditions as the production conditions (welding conditions) of the welded joint, and measuring the nugget diameter at each boundary level between the k th steel sheet and the $(k+1)$ th steel sheet in the sample. The amount of nugget enlargement can be adjusted, for example, by preparing in advance the same sheet combination as the sheet combination to be welded and conducting preliminary welding tests under various conditions within the above welding condition ranges to investigate the amount of nugget enlargement under each condition.

[0249] Conditions other than those described above are not particularly limited, and a conventional method may be used. For example, the electrode force may be 1.5 kN to 10.0 kN. The electrode force may be the same or a different value than in the first current process. The current value during current passing may be constant or may be varied as appropriate during current passing, such as in upslope current, as long as the current value is 11 or more.

Pressure Hold Process

[0250] After the second current process described above, the pressure hold process is performed, in which pressure is held on the sheet combination in a no-current state.

Pressure Hold Time: 10 ms or More

[0251] As described above, the second current process enlarges the nugget as described above, and therefore the nugget is in a molten state at the end of the second current process. Therefore, when releasing the electrode from the sheet combination immediately after the second current passage process, cracks may occur from the nugget end or corona bond end between steel sheets of the sheet combination toward the inside of the nugget, resulting in a decrease in joint quality. Such cracks are more pronounced in a sheet combination of material to be joined where a boundary between the steel sheets of the sheet combination is located away from the center position ($1/2$ thickness position of the sheet combination) of the overall sheet combination thickness. Examples of such sheet combinations include sheet combinations that consist of two steel sheets of different thicknesses stacked on top of each other, and sheet combinations that consist of three or more steel sheets stacked on top of each other, regardless of their

thickness. From the viewpoint of effectively preventing the occurrence of such cracks regardless of the sheet combination, after the second current process, the sheet combination is held under pressure in the no-current state, and the pressure hold time is 10 ms or more. The pressure hold time is preferably 20 ms or more. An upper limit of the pressure hold time is not particularly limited. From the viewpoint of production efficiency, the pressure hold time is preferably 2000 ms or less.

[0252] The electrode force in the pressure hold process is not particularly limited. For example, the electrode force may be 1.5 kN to 10.0 kN. The electrode force in the pressure hold process may be the same as or different from that of the first current process and the second current process.

[0253] Conditions other than those described above are not particularly limited, and a conventional method may be used.

[0254] Further, the resistance spot welding method according to an embodiment of the present disclosure obtains a nugget diameter of the desired size while suppressing the occurrence of cracking in the welded portion regardless of the influence of a disturbance and regardless of sheet combination. Therefore, the resistance spot welding method according to an embodiment of the present disclosure is particularly suitable when applied to difficult welding conditions, for example, one or more of the following conditions (A) to (E):

[0255] (A) a welding electrode has an inclined angle relative to the sheet combination,

[0256] (B) the pair of welding electrodes are off-center,

[0257] (C) before pressing the sheet combination, there is a gap between the fixed welding electrode and the sheet combination (hereinafter also referred to as an electrode-sheet combination gap),

[0258] (D) before pressing the sheet combination, there is at least one gap between the steel sheets of the sheet combination (hereinafter also referred to as a sheet gap), and

[0259] (E) on the surface of the sheet combination, the shortest distance from the center of the welding contact point to an end face of the sheet combination (hereinafter also referred to as a shortest end face distance) is 10 mm or less.

[0260] Here, the “inclined angle” in (A) is the angle of the axis of the welding electrode relative to the perpendicular direction to the surface of the sheet combination. The “welding electrode has an inclined angle relative to the sheet combination” means a state in which the inclined angle is not 0° , that is, the perpendicular direction to the surface of the sheet combination and the axis of the welding electrode (at least one of the axis of the upper electrode or the axis of the lower electrode) are not parallel to each other.

[0261] The “pair of welding electrodes are off-center” in (B) means a state in which the axis of the upper electrode and the axis of the lower electrode of the welding electrodes are not aligned. Further, the amount of misalignment is the distance between the axis of the upper electrode and the axis of the lower electrode of the welding electrodes.

[0262] The “before pressing the sheet combination” in (C) and (D) means after the sheet combination is placed in a welding device including a fixed welding electrode (corresponding to the lower electrode as one example) and a driven welding electrode (corresponding to the upper elec-

trode as one example), and before the driven welding electrode is moved to start pressing the sheet combination. [0263] The resistance spot welding method according to an embodiment of the present disclosure is applicable not only to a sheet combination of two steel sheets overlapped, but also to a sheet combination of three or more steel sheets overlapped.

[4] Method of Producing Welded Member

[0264] The method of producing a welded member according to an embodiment of the present disclosure includes the process of joining a sheet combination by the resistance spot welding method. This allows a stable nugget diameter of the desired size to be secured while suppressing deformation of the shoulders of the sheet combination and therefore suppressing cracking of the welded portion. As a result, it is possible to produce various welded members with high production efficiency, in particular automotive parts and the like that include a galvanized steel sheet having high corrosion resistance on an outermost surface.

Examples

[0265] Resistance spot welding was performed on the sheet combinations listed in Table 1 under the conditions listed in Table 2 to produce welded joints. For each sample number, disturbances (welding conditions (A) to (E) as described above) were simulated as listed in Table 1. In Table 1, “—” in columns (A), (B), (C), and (E) means that the corresponding condition was not satisfied. Further, in a sheet combination simulating a sheet gap of (D), for example, three 30 mm×100 mm steel sheets were stacked on top of each other and 30 mm×25 mm spacers 7, 8 were placed between the second and third steel sheets to provide a 2 mm sheet gap, as illustrated in FIG. 3. Note that “0 mm” in the sheet gap column of (D) means no sheet gap. Further, in Table 1, “—” in column (E) means that the welded portion (nugget) was formed so that the center of the 30 mm×100 mm steel sheet was the center of the weld contact point. Further, the first current process was performed with an upslope current, except for sample number 2. For sample number 2, current was passed at a constant current value. The second current process was performed in the order of cooling and then current passing, and each cycle of cooling and current passing was performed under the same conditions.

[0266] Further, resistance spot welding was performed at room temperature and with the welding electrode always being water cooled. Both the upper electrode (driven welding electrode) and the lower electrode (fixed welding electrode) used a DR-type electrode made of chromium copper having a tip diameter of 6 mm and a curvature radius of 40 mm. Further, the electrode force was controlled by driving

the upper electrode with a servo motor, and DC power was supplied during current passing. For all sample numbers, the sheet combinations were arranged so that the first steel sheet came into contact with the upper electrode (driven welding electrode).

[0267] Further, for the welded joints obtained, by the methods described above, the indentation ratio b/T of the shoulders of the sheet combination was measured and the nugget diameter x_1 at the boundary level between the first steel sheet and the second steel sheet was measured. For sheet combinations of three steel sheets, the nugget diameter x_2 at the boundary level between the second steel sheet and the third steel sheet was measured. For sheet combinations of four steel sheets, the nugget diameter x_3 at the boundary level between the third steel sheet and the fourth steel sheet was measured. The measurement results are listed in Table 3.

[0268] Further, surfaces and cross-sections of the nuggets of the welded joints were observed to visually check for the presence of shoulder cracking and cracking from between the steel sheets towards the interior of the nuggets. The results are listed in Table 3.

[0269] In Table 3, A, B, and C in the shoulder cracking column mean the following, respectively.

[0270] A (pass, especially good): no cracking

[0271] B (pass): cracks occurred, but length was less than 50 μ m

[0272] C (fail): cracks of 50 μ m or more in length occurred

[0273] The evaluation column in Table 3 is labeled “Pass” when the shoulder cracking evaluation was A or B, there was no crack from the steel sheets towards the interior of the nugget, and the nugget diameter x_k was $3.5\sqrt{t_k}$ or more, and “Fail” when any of the above cases was not true, that is, one of the target properties was not obtained.

[0274] Further, additional samples were separately prepared under the same conditions as each sample number, but with current applied only up to the first current process. Further, for these samples, the nugget diameter x_1' at the boundary level between the first steel sheet and the second steel sheet was measured. For sheet combinations of three steel sheets, the nugget diameter x_2' at the boundary level between the second steel sheet and the third steel sheet was measured. For sheet combinations of four steel sheets, the nugget diameter x_3' at the boundary level between the third steel sheet and the fourth steel sheet was measured. Then, x_1-x_1' , x_2-x_2' , and x_3-x_3' were calculated. Here, when the maximum value of these values was 0.05 mm or more, the column “Enlargement of nugget in second current process” in Table 2 is labeled “Yes”, and when the maximum value of these values was less than 0.05 mm, the column is labeled “No”.

TABLE 1

Sheet combination															
Sample No.	1st steel sheet				2nd steel sheet				3rd steel sheet				4th steel sheet		
	Ref. sign	Strength MPa	Type	Thickness mm	Ref. sign	Strength MPa	Type	Thickness mm	Ref. sign	Strength MPa	Type	Thickness mm	Ref. sign	Strength MPa	Type
1	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
2	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
3	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—

TABLE 1-continued

4	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
5	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
6	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
7	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
8	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
9	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
10	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
11	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
12	A	270	GA	0.7	B	1470	GA	1.2	C	1470	GA	1.4	—	—	—
13	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
14	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
15	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
16	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
17	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
18	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
19	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
20	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
21	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
22	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
23	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
24	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
25	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
26	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
27	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
28	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
29	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
30	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
31	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
32	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
33	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
34	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
35	D	270	GA	0.6	E	980	CR	1.3	F	980	GA	1.0	G	1470	GA
36	H	980	CR	1.1	I	270	CR	1.1	J	1800	CR	1.2	K	1470	GA
37	H	980	CR	1.1	I	270	CR	1.1	J	1800	CR	1.2	K	1470	GA
38	H	980	CR	1.1	I	270	CR	1.1	J	1800	CR	1.2	K	1470	GA
39	H	980	CR	1.1	I	270	CR	1.1	J	1800	CR	1.2	K	1470	GA
40	H	980	CR	1.1	I	270	CR	1.1	J	1800	CR	1.2	K	1470	GA
41	L	1800	GA	1.2	M	1470	GA	1.8	—	—	—	—	—	—	—
42	L	1800	GA	1.2	M	1470	GA	1.8	—	—	—	—	—	—	—
43	N	270	GI	0.6	O	980	CR	1.3	F	980	GI	1.0	P	1470	GI
44	N	270	GI	0.6	O	980	CR	1.3	F	980	GI	1.0	P	1470	GI
45	N	270	GI	0.6	O	980	CR	1.3	F	980	GI	1.0	P	1470	GI
46	Q	270	GF	0.6	R	980	CR	1.3	F	980	GF	1.0	S	1470	GF
47	Q	270	GF	0.6	R	980	CR	1.3	F	980	GF	1.0	S	1470	GF
48	Q	270	GF	0.6	R	980	CR	1.3	F	980	GF	1.0	S	1470	GF
49	T	270	Ecogal	0.6	U	980	CR	1.3	F	980	Ecogal	1.0	V	1470	Ecogal
50	T	270	Ecogal	0.6	U	980	CR	1.3	F	980	Ecogal	1.0	V	1470	Ecogal
51	T	270	Ecogal	0.6	U	980	CR	1.3	F	980	Ecogal	1.0	V	1470	Ecogal
52	W	270	GL	0.6	X	980	CR	1.3	F	980	GL	1.0	Y	1470	GL
53	W	270	GL	0.6	X	980	CR	1.3	F	980	GL	1.0	Y	1470	GL
54	W	270	GL	0.6	X	980	CR	1.3	F	980	GL	1.0	Y	1470	GL

Disturbance																
										(C)		(D)			(E)	
										Electrode-sheet		1st-2nd	2nd-3rd	3rd-4th	Shortest distance	
												sheet gap mm	sheet gap mm	sheet gap mm	to end face mm	
												gap mm	gap mm	gap mm		
Sample No.	Sheet combination				(A)		(B)									
	4th steel sheet	n	T	T/t _U	T/t _L	Inclination angle °	Misalignment mm	combination	sheet gap mm	sheet gap mm	sheet gap mm	sheet gap mm	sheet gap mm	sheet gap mm	sheet gap mm	sheet gap mm
	Thickness mm	Sheets	mm													
1	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
2	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
3	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
4	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
5	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
6	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
7	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
8	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
9	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
10	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
11	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
12	—	3	3.3	4.71	2.36	—	—	—	0	2	—	—	—	—	—	—
13	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—	—	—	—	—
14	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—	—	—	—	—

TABLE 1-continued

15	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
16	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
17	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
18	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
19	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
20	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
21	1.2	4	4.1	6.83	3.42	4	1	2	0	2	0	—
22	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	5
23	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
24	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
25	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
26	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
27	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
28	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
29	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
30	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
31	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
32	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
33	1.2	4	4.1	6.83	3.42	4	1	2	0	2	0	—
34	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	5
35	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	5
36	1.3	4	4.7	4.27	3.62	—	—	—	0	0	0	—
37	1.3	4	4.7	4.27	3.62	—	—	—	0	0	0	—
38	1.3	1	4.7	4.27	3.62	—	—	—	0	0	0	—
39	1.3	1	4.7	4.27	3.62	—	—	—	0	0	0	—
40	1.3	4	4.7	4.27	3.62	—	—	—	0	0	0	—
41	—	2	3.0	2.50	1.67	—	—	—	0	0	0	—
42	—	2	3.0	2.50	1.67	—	—	—	0	0	0	—
43	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
44	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
45	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
46	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
47	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
48	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
49	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
50	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
51	1.2	1	4.1	6.83	3.42	4	—	2	0	2	0	—
52	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
53	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—
54	1.2	4	4.1	6.83	3.42	4	—	2	0	2	0	—

GA: galvanized steel sheet

GI: hot-dip galvanized steel sheet

GF: Galfan steel sheet

GL: Galvalume steel sheet

CR: Steel sheet without coating (cold-rolled steel sheet)

TABLE 2

First current process													Second current process			
		Current value		Weld time						Current passing					Cooling and current passing cycles No.	
Sample No.	Electrode force kN	Upslope start	Upslope end	I1 kA	α ms	Expression (4) lower	Expression (5) lower	Expulsion occurrence	Cooling time ms	Current value kA	Weld time ms					
		current Is kA	current If kA			limit (1st sheet) ms	limit (nth sheet) ms									
1	3.5	4.5	6.0	6.0	200	108.6	76.8	No	20	8.0	40	4				
2	3.5	—	—	5.0	200	108.6	76.8	No	20	8.0	40	4				
3	3.5	5.5	7.0	7.0	200	108.6	76.8	No	20	10.0	40	4				
4	3.5	6.5	8.0	8.0	200	108.6	76.8	No	10	8.0	15	4				
5	3.5	6.5	8.0	8.0	200	108.6	76.8	No	60	9.0	40	4				
6	3.5	6.5	8.0	8.0	200	108.6	76.8	No	60	8.0	40	4				
7	3.5	6.5	7.0	7.0	60	108.6	76.8	No	80	9.0	40	4				
8	3.5	6.5	8.0	8.0	200	108.6	76.8	No	20	8.5	140	4				
9	3.5	6.5	8.0	8.0	140	108.6	76.8	No	100	9.0	20	4				
10	3.5	6.5	8.0	8.0	140	108.6	76.8	No	100	9.0	20	4				
11	3.5	6.5	8.0	8.0	200	108.6	76.8	No	10	8.0	10	4				
12	3.5	6.5	8.0	8.0	200	108.6	76.8	No	5	8.0	15	4				
13	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	9.5	40	3				
14	3.5	7.5	8.5	8.5	200	130.7	92.4	Yes	20	9.0	40	3				
15	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	10.5	40	3				

TABLE 2-continued

16	3.5	6.5	8.0	8.0	200	130.7	92.4	No	60	13.0	20	3
17	3.5	7.0	8.5	8.5	200	130.7	92.4	No	60	9.5	40	3
18	3.5	7.0	8.0	8.0	140	130.7	92.4	No	20	12.0	40	3
19	3.5	6.5	8.0	8.0	200	130.7	92.4	No	10	9.5	15	3
20	3.5	6.5	8.0	8.0	200	130.7	92.4	No	40	9.0	140	3
21	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	10.5	40	3
22	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	10.5	40	3
23	3.5	6.5	8.0	8.0	200	130.7	92.4	No	—	—	—	—
24	3.5	7.0	8.0	8.0	120	130.7	92.4	No	20	12.0	40	3
25	3.5	8.0	9.0	9.0	120	130.7	92.4	Yes	20	12.0	40	3
26	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	9.0	120	3
27	3.5	6.5	8.0	8.0	200	130.7	92.4	No	120	13.0	20	3
28	3.5	6.5	8.0	8.0	200	130.7	92.4	No	120	14.0	20	3
29	3.5	7.0	8.5	8.5	200	130.7	92.4	No	60	8.5	40	3
30	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	9.0	120	3
31	3.5	6.5	8.0	8.0	200	130.7	92.4	No	10	9.5	10	3
32	3.5	6.5	8.0	8.0	200	130.7	92.4	No	5	9.5	15	3
33	3.5	7.0	8.0	8.0	80	130.7	92.4	No	20	13.0	40	3
34	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	10.5	120	3
35	3.5	6.5	8.0	8.0	200	130.7	92.4	No	120	10.5	20	3
36	3.5	6.5	8.0	8.0	200	—	95.1	No	20	9.0	40	4
37	3.5	6.5	8.0	8.0	200	—	95.1	No	20	10.5	40	4
38	3.5	6.5	8.0	8.0	60	—	95.1	No	20	10.5	40	4
39	3.5	6.5	8.0	8.0	200	—	95.1	No	20	10.5	120	4
40	3.5	6.5	8.0	8.0	200	—	95.1	No	100	10.5	20	4
41	3.5	5.0	6.5	6.5	140	79.1	64.5	No	20	9.0	100	2
42	3.5	5.0	6.5	6.5	140	79.1	64.5	No	10	9.0	100	2
43	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	10.5	40	3
44	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	9.0	120	3
45	3.5	6.5	8.0	8.0	200	130.7	92.4	No	120	13.0	20	3
46	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	10.5	40	3
47	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	9.0	120	3
48	3.5	6.5	8.0	8.0	200	130.7	92.4	No	120	13.0	20	3
49	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	10.5	40	3
50	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	9.0	120	3
51	3.5	6.5	8.0	8.0	200	130.7	92.4	No	120	13.0	20	3
52	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	10.5	40	3
53	3.5	6.5	8.0	8.0	200	130.7	92.4	No	20	9.0	120	3
54	3.5	6.5	8.0	8.0	200	130.7	92.4	No	120	13.0	20	3

Sample No.	Second current process						Pressure		
	Ratio of total cooling time β to total weld time γ					Enlargement	hold		Remarks
	β/γ	Expression (6) lower limit (1st sheet)	Expression (6) upper limit (1st sheet)	Expression (7) lower limit (nth sheet)	Expression (7) upper limit (nth sheet)		process Pressure ms	hold time ms	
1	0.50	0.22	4.34	0.15	3.07	Yes	No	15	Example
2	0.50	0.22	4.34	0.15	3.07	Yes	No	20	Example
3	0.50	0.22	4.34	0.15	3.07	Yes	Yes	40	Example
4	0.67	0.22	4.34	0.15	3.07	Yes	Yes	60	Example
5	1.50	0.22	4.34	0.15	3.07	Yes	No	40	Example
6	1.50	0.22	4.34	0.15	3.07	No	No	40	Comparative Example
7	2.00	0.22	4.34	0.15	3.07	Yes	Yes	40	Comparative Example
8	0.14	0.22	4.34	0.15	3.07	Yes	Yes	20	Comparative Example
9	5.00	0.22	4.34	0.15	3.07	No	No	60	Comparative Example
10	5.00	0.22	4.34	0.15	3.07	No	No	5	Comparative Example
11	1.00	0.22	4.34	0.15	3.07	No	No	60	Comparative Example
12	0.33	0.22	4.34	0.15	3.07	Yes	Yes	60	Comparative Example
13	0.50	0.26	5.23	0.18	3.70	Yes	No	15	Example
14	0.50	0.26	5.23	0.18	3.70	Yes	No	40	Example
15	0.50	0.26	5.23	0.18	3.70	Yes	Yes	20	Example
16	3.00	0.26	5.23	0.18	3.70	Yes	Yes	100	Example
17	1.50	0.26	5.23	0.18	3.70	Yes	No	80	Example
18	0.50	0.26	5.23	0.18	3.70	Yes	Yes	80	Example
19	0.67	0.26	5.23	0.18	3.70	Yes	Yes	80	Example
20	0.29	0.26	5.23	0.18	3.70	Yes	Yes	40	Example
21	0.50	0.26	5.23	0.18	3.70	Yes	Yes	60	Example
22	0.50	0.26	5.23	0.18	3.70	Yes	Yes	80	Example
23	—	—	—	—	—	—	—	—	Comparative Example
24	0.50	0.26	5.23	0.18	3.70	Yes	Yes	60	Comparative Example
25	0.50	0.26	5.23	0.18	3.70	Yes	Yes	40	Comparative Example
26	0.17	0.26	5.23	0.18	3.70	Yes	Yes	20	Comparative Example
27	6.00	0.26	5.23	0.18	3.70	No	No	40	Comparative Example
28	6.00	0.26	5.23	0.18	3.70	Yes	No	80	Comparative Example

TABLE 2-continued

29	1.50	0.26	5.23	0.18	3.70	No	No	80	Comparative Example
30	0.17	0.26	5.23	0.18	3.70	Yes	Yes	5	Comparative Example
31	1.00	0.26	5.23	0.18	3.70	No	No	80	Comparative Example
32	0.33	0.26	5.23	0.18	3.70	Yes	Yes	80	Comparative Example
33	0.50	0.26	5.23	0.18	3.70	Yes	Yes	100	Comparative Example
34	0.17	0.26	5.23	0.18	3.70	Yes	Yes	40	Comparative Example
35	6.00	0.26	5.23	0.18	3.70	No	Yes	15	Comparative Example
36	0.50	—	—	0.19	3.80	Yes	No	100	Example
37	0.50	—	—	0.19	3.80	Yes	Yes	100	Example
38	0.50	—	—	0.19	3.80	Yes	Yes	100	Comparative Example
39	0.17	—	—	0.19	3.80	Yes	Yes	100	Comparative Example
40	5.00	—	—	0.19	3.80	No	No	100	Comparative Example
41	0.20	0.16	3.16	0.13	2.58	Yes	Yes	120	Example
42	0.10	0.16	3.16	0.13	2.58	Yes	Yes	120	Comparative Example
43	0.50	0.26	5.23	0.18	3.70	Yes	Yes	20	Example
44	0.17	0.26	5.23	0.18	3.70	Yes	Yes	20	Comparative Example
45	6.00	0.26	5.23	0.18	3.70	No	No	40	Comparative Example
46	0.50	0.26	5.23	0.18	3.70	Yes	Yes	20	Example
47	0.17	0.26	5.23	0.18	3.70	Yes	Yes	20	Comparative Example
48	6.00	0.26	5.23	0.18	3.70	No	No	40	Comparative Example
49	0.50	0.26	5.23	0.18	3.70	Yes	Yes	20	Example
50	0.17	0.26	5.23	0.18	3.70	Yes	Yes	20	Comparative Example
51	6.00	0.26	5.23	0.18	3.70	No	No	40	Comparative Example
52	0.50	0.26	5.23	0.18	3.70	Yes	Yes	20	Example
53	0.17	0.26	5.23	0.18	3.70	Yes	Yes	20	Comparative Example
54	6.00	0.26	5.23	0.18	3.70	No	No	40	Comparative Example

TABLE 3

Welded portion cracking										
Shoulder indentation ratio			Crack from between							
Expression		Expression	Shoulder cracking			steel sheets				
(1) upper		(2) upper	Nugget diameter			1st	nth	toward		
Sample No.	b/T	limit (1st sheet)	limit (nth sheet)	x ₁ xv ₁ t ₁	x ₂ xv ₂ t ₂	x ₃ xv ₃ t ₃	sheet surface	sheet surface	nugget interior	Evaluation Remarks
1	0.120	0.526	0.319	3.9	5.3	—	A	A	No	Pass Example
2	0.180	0.526	0.319	3.5	5.1	—	A	B	No	Pass Example
3	0.270	0.526	0.319	4.5	5.5	—	A	B	No	Pass Example
4	0.190	0.526	0.319	3.7	5.2	—	A	B	No	Pass Example
5	0.140	0.526	0.319	3.5	5.2	—	A	A	No	Pass Example
6	0.090	0.526	0.319	3.4	4.9	—	A	A	No	Fail Comparative Example
7	0.090	0.526	0.319	2.8	3.4	—	A	A	No	Fail Comparative Example
8	0.320	0.526	0.319	5.3	5.1	—	A	C	No	Fail Comparative Example
9	0.100	0.526	0.319	3.2	4.3	—	A	A	No	Fail Comparative Example
10	0.100	0.526	0.319	3.2	4.3	—	A	A	Yes	Fail Comparative Example
11	0.090	0.526	0.319	3.4	4.9	—	A	A	No	Fail Comparative Example
12	0.320	0.526	0.319	3.5	5.0	—	A	C	No	Fail Comparative Example
13	0.080	0.437	0.265	4.6	5.7	5.4	A	A	No	Pass Example
14	0.150	0.437	0.265	4.5	5.6	5.3	A	A	No	Pass Example
15	0.170	0.437	0.265	4.9	6.0	5.7	A	B	No	Pass Example
16	0.230	0.437	0.265	3.6	5.4	5.1	A	B	No	Pass Example
17	0.100	0.437	0.265	3.5	5.5	5.0	A	A	No	Pass Example
18	0.190	0.437	0.265	3.6	5.3	4.8	A	B	No	Pass Example
19	0.180	0.437	0.265	3.6	5.5	5.0	A	B	No	Pass Example
20	0.240	0.437	0.265	5.8	6.3	5.9	A	B	No	Pass Example
21	0.190	0.437	0.265	4.8	5.8	5.3	A	B	No	Pass Example
22	0.180	0.437	0.265	4.9	5.9	5.5	A	B	No	Pass Example
23	0.070	0.437	0.265	2.9	4.9	4.5	A	A	No	Fail Comparative Example
24	0.170	0.437	0.265	3.3	5.1	4.7	A	B	No	Fail Comparative Example
25	0.270	0.437	0.265	3.3	5.0	4.7	A	C	No	Fail Comparative Example
26	0.300	0.437	0.265	6.6	6.8	6.3	A	C	No	Fail Comparative Example
27	0.070	0.437	0.265	2.9	4.9	4.5	A	A	No	Fail Comparative Example
28	0.270	0.437	0.265	3.3	5.3	4.9	A	C	No	Fail Comparative Example
29	0.090	0.437	0.265	3.3	5.3	4.8	A	A	No	Fail Comparative Example
30	0.300	0.437	0.265	6.6	6.8	6.3	A	C	Yes	Fail Comparative Example
31	0.070	0.437	0.265	2.9	4.9	4.5	A	A	No	Fail Comparative Example
32	0.270	0.437	0.265	3.5	5.2	4.9	A	C	No	Fail Comparative Example
33	0.280	0.437	0.265	3.4	5.3	4.8	A	C	No	Fail Comparative Example

TABLE 3-continued

Welded portion cracking											
Shoulder indentation ratio				Crack from between							
Sample No.	b/T	Expression	Expression	Shoulder cracking			steel sheets			Evaluation	Remarks
		(1) upper	(2) upper	Nugget diameter			1st	nth	toward		
		limit (1st sheet)	limit (nth sheet)	x ₁ x√t ₁	x ₂ x√t ₂	x ₃ x√t ₃	sheet surface	sheet surface	nugget interior		
34	0.310	0.437	0.265	6.1	6.9	6.5	A	C	No	Fail	Comparative Example
35	0.070	0.437	0.265	2.9	4.9	4.5	A	A	No	Fail	Comparative Example
36	0.090	—	0.258	4.3	5.2	5.0	A	A	No	Pass	Example
37	0.180	—	0.258	4.6	5.6	5.5	A	B	No	Pass	Example
38	0.150	—	0.258	2.7	3.8	3.3	A	B	No	Fail	Comparative Example
39	0.280	—	0.258	4.8	5.9	5.7	A	C	No	Fail	Comparative Example
40	0.080	—	0.258	3.3	4.0	3.8	A	A	No	Fail	Comparative Example
41	0.240	0.280	0.379	5.4	—	—	B	A	No	Pass	Example
42	0.300	0.280	0.379	6.0	—	—	C	B	No	Fail	Comparative Example
43	0.160	0.437	0.265	4.8	5.9	5.7	A	B	No	Pass	Example
44	0.290	0.437	0.265	6.5	6.9	6.2	A	C	No	Fail	Comparative Example
45	0.060	0.437	0.265	2.8	4.8	4.5	A	A	No	Fail	Comparative Example
46	0.190	0.437	0.265	5.0	6.0	5.8	A	B	No	Pass	Example
47	0.300	0.437	0.265	6.7	6.9	6.3	A	C	No	Fail	Comparative Example
48	0.080	0.437	0.265	3.0	4.9	4.5	A	A	No	Fail	Comparative Example
49	0.180	0.437	0.265	4.9	6.1	5.7	A	B	No	Pass	Example
50	0.310	0.437	0.265	6.6	6.8	6.4	A	C	No	Fail	Comparative Example
51	0.080	0.437	0.265	2.9	5.0	4.6	A	A	No	Fail	Comparative Example
52	0.170	0.437	0.265	4.8	5.8	5.6	A	B	No	Pass	Example
53	0.280	0.437	0.265	6.4	6.7	6.3	A	C	No	Fail	Comparative Example
54	0.070	0.437	0.265	2.8	4.7	4.4	A	A	No	Fail	Comparative Example

[0275] According to Table 3, for all of the Examples, nugget diameters of the desired size were obtained while suppressing the occurrence of cracking in the welded portion.

[0276] In contrast, for the Comparative Examples having sample numbers 6, 9, 10, 11, 27, 29, 31, 35, 40, 45, 48, 51, and 54, the nuggets did not enlarge sufficiently in the second current process to obtain the desired size of nugget diameter.

[0277] For sample numbers 7, 24, 25, 33, and 38, weld time in the first current process was insufficient, and therefore the desired size of nugget diameter was not obtained. When the weld time of the first current process was insufficient, and the current value of the first current process was increased to increase the nugget diameter, nuggets of the desired size were not obtained, as in sample numbers 25 and 33. At the same time, deformation of shoulders of the sheet combination became excessive, and therefore the shoulder indentation ratio became excessive, and shoulder cracking became more pronounced.

[0278] For sample numbers 8, 26, 30, 34, 39, 42, 44, 47, 50, and 53, the ratio of cooling time in the second current process was too small and β/γ did not satisfy the defined range. As a result, the amount of expulsion (the amount of spattered molten metal) could not be sufficiently controlled in the second current process, resulting in excessive deformation of the shoulders of the sheet combination and, consequently, excessive shoulder indentation ratio, and shoulder cracking became more pronounced.

[0279] For sample numbers 9, 10, 27, 28, 35, 40, 45, 48, 51, and 54, the ratio of cooling time in the second current process was too large and β/γ did not satisfy the defined range. As a result, the nuggets did not enlarge sufficiently to obtain the desired size of nugget diameter. When the ratio of cooling time in the second current process was too large and

the current value of the second current process was increased to increase the nugget diameter, a nugget of the desired size was not obtained, as in sample number 28. At the same time, deformation of shoulders of the sheet combination became excessive, and therefore the shoulder indentation ratio became excessive, and shoulder cracking became more pronounced.

[0280] For sample numbers 10 and 30, the pressure hold time was too short, and therefore cracks occurred between the steel sheets towards the nugget interior.

[0281] For sample numbers 11 and 31, weld time per cycle in the second current process was insufficient, and therefore the nuggets did not enlarge sufficiently, and the desired size of nugget diameter was not obtained.

[0282] For sample numbers 12 and 32, cooling time per cycle in the second current process was insufficient, and therefore deformation of shoulders of the sheet combination became excessive and therefore the shoulder indentation ratio became excessive, and shoulder cracking became more pronounced.

[0283] For sample number 23, the second current process was not performed, that is, only one stage of current passing was performed, and the desired size of nugget diameter was not obtained.

REFERENCE SIGNS LIST

- [0284] 1-1 steel sheet (upper steel sheet)
- [0285] 1-2 steel sheet (lower steel sheet)
- [0286] 1-3 steel sheet
- [0287] 2 sheet combination
- [0288] 3 welding electrode (upper electrode)
- [0289] 4 welding electrode (lower electrode)
- [0290] 5 nugget

[0291] 6 shoulder

[0292] 7 spacer

[0293] 8 spacer

1. A welded joint comprising a sheet combination of n overlapped steel sheets and a nugget joining the steel sheets, wherein

n is an integer greater than or equal to 2,

in the sheet combination, at least one of the first steel sheet or the nth steel sheet in order from the top is a galvanized steel sheet,

an indentation ratio b/T of shoulders of the sheet combination satisfies,

in the case of [Condition 1], the following Expression (1),

in the case of [Condition 2], the following Expression (2),

in the case of [Condition 3], Expressions (1) and (2),

in the sheet combination, nugget diameter x_k , in mm, at each boundary level between a kth steel sheet and a (k+1)th steel sheet is $3.5\sqrt{t_k}$ or more, k is an integer from 1 to n-1, t_k is the thickness in mm of the thinner of the kth steel sheet and the (k+1)th steel sheet,

$$b/T \leq 0.6(T/t_U)^{0.5} \times (980/S_U)^{0.5} \quad (1)$$

$$b/T \leq 0.6(T/t_L)^{0.5} \times (980/S_L)^{0.5} \quad (2)$$

where

b is shoulder indentation in mm,

T is total thickness in mm of the n steel sheets,

t_U is thickness in mm of the first steel sheet,

t_L is thickness in mm of the nth steel sheet,

S_U is tensile strength in MPa of the first steel sheet,

S_L is tensile strength in MPa of the nth steel sheet,

the shoulder indentation b is determined by the following

Expression (3),

$$b = T - a \quad (3)$$

where

a is minimum thickness in mm of the sheet combination at a distance of 1 mm from a nugget end,

and [Condition 1] to [Condition 3] are as follows:

[Condition 1]

of the first steel sheet and the nth steel sheet of the sheet combination, only the first steel sheet is a galvanized steel sheet,

[Condition 2]

of the first steel sheet and the nth steel sheet of the sheet combination, only the nth steel sheet is a galvanized steel sheet,

[Condition 3]

of the first steel sheet and the nth steel sheet of the sheet combination, both the first steel sheet and the nth steel sheet are galvanized steel sheets.

2. A welded member comprising the welded joint according to claim 1.

3. A resistance spot welding method of squeezing, by a pair of welding electrodes, a sheet combination of n overlapped steel sheets and passing a current while applying an electrode force to join the sheet combination, wherein

n is an integer greater than or equal to 2,

in the sheet combination, at least one of the first steel sheet or the nth steel sheet in order from the top is a galvanized steel sheet,

the resistance spot welding method comprising:

a first current process of forming a nugget;

a second current process of enlarging the nugget formed in the first current passage process; and

after the second current process, a pressure hold process of holding pressure on the sheet combination in a no-current state,

wherein,

in the first current process, current is passed for a weld time of α ms,

the weld time α , in ms, satisfies,

in the case of [Condition 1], the following Expression (4),

in the case of [Condition 2], the following Expression (5),

in the case of [Condition 3], Expressions (4) and (5),

the second current process comprises:

cooling where no current is passed for a cooling time of 10 ms or more; and

current passing for a weld time of 15 ms or more with a current value greater than or equal to a current value of the first current process, the cooling and the current passing each being carried out at least once, wherein a ratio β/γ of total cooling time β to total weld time γ in the second current process satisfies,

in the case of [Condition 1], Expression (6),

in the case of [Condition 2], Expression (7),

in the case of [Condition 3], Expressions (6) and (7),

in the pressure hold process, the pressure hold time is 10 ms or more,

$$\alpha \geq 50 \times (T/t_U)^{0.5} \quad (4)$$

$$\alpha \geq 50 \times (T/t_L)^{0.5} \quad (5)$$

$$0.10 \times (T/t_U)^{0.5} \leq \beta/\gamma \leq 2 \times (T/t_U)^{0.5} \quad (6)$$

$$0.10 \times (T/t_L)^{0.5} \leq \beta/\gamma \leq 2 \times (T/t_L)^{0.5} \quad (7)$$

where

T is total thickness in mm of the n steel sheets,

t_U is thickness in mm of the first steel sheet,

t_L is thickness in mm of the nth steel sheet,

and [Condition 1] to [Condition 3] are as follows:

[Condition 1]

of the first steel sheet and the nth steel sheet of the sheet combination, only the first steel sheet is a galvanized steel sheet,

[Condition 2]

of the first steel sheet and the nth steel sheet of the sheet combination, only the nth steel sheet is a galvanized steel sheet,

[Condition 3]

of the first steel sheet and the nth steel sheet of the sheet combination, both the first steel sheet and the nth steel sheet are galvanized steel sheets.

4. The resistance spot welding method according to claim 3, satisfying one or more of the following conditions (A) to (E),

(A) a welding electrode has an inclined angle relative to the sheet combination,

(B) the pair of welding electrodes are off-center,

(C) before pressing the sheet combination, there is a gap between the fixed welding electrode and the sheet combination,

(D) before pressing the sheet combination, there is at least one gap between the steel sheets of the sheet combination, and

(E) on the surface of the sheet combination, the shortest distance from the center of the welding contact point to an end face of the sheet combination is 10 mm or less.

5. A method of producing a welded member, comprising a process of joining a sheet combination of n overlapped steel sheets by the resistance spot welding method according to claim 3, wherein n is an integer greater than or equal to 2.

6. A method of producing a welded member, comprising a process of joining a sheet combination of n overlapped steel sheets by the resistance spot welding method according to claim 4, wherein n is an integer greater than or equal to 2.

* * * * *